A narrow-band DG level-set method on cut cells

semi-Lagrangian advection and elliptic reinitialization

MeltPoolDG – application `mp-narrow-band`

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Abstract

The level set is represented by discontinuous Galerkin (DG) polynomials only on the cells cut by the interface. It is advected with a semi-Lagrangian method: the new nodal values are obtained by tracing the characteristics back to the old cut cells, without any extension of the level set into neighbor cells. Cells that are not cut store only the sign of the level set. After each step, the level set is reinitialized on the new cut cells by an elliptic, minimization-based method with a symmetric interior penalty (SIPG) discretization. This note summarizes the scheme, gives the algorithms and reports results for a rotating circle and the single-vortex test.

1 Overview

The level-set field ϕ describes the interface $\Gamma = \{\phi = 0\}$; in this note $\phi > 0$ inside the bubble. The method keeps three sets of cells of a Cartesian mesh $\mathcal{T}$ (Figure 1):

- the *cut cells* $\mathcal{T}^\Gamma$, which contain the discrete interface $\Gamma_h = \{\phi_h = 0\}$; only these cells carry a polynomial $\phi_h|_K \in Q_p(K)$ (`FE_DGQ`),
- the *candidate cells* $\mathcal{T}^C \supset \mathcal{T}^\Gamma$, the cut cells and the cells sharing a vertex with a cut cell; they hold the new level set during one time step,
- all other cells, which store only a sign $s_K \in \{-1, +1\}$.

A time step consists of three stages: the semi-Lagrangian evaluation of the new level set on the candidate cells (Section 3), the detection of the new cut cells (Section 2) and the elliptic reinitialization on the new cut cells (Section 4). The complete step is given in Section 5.

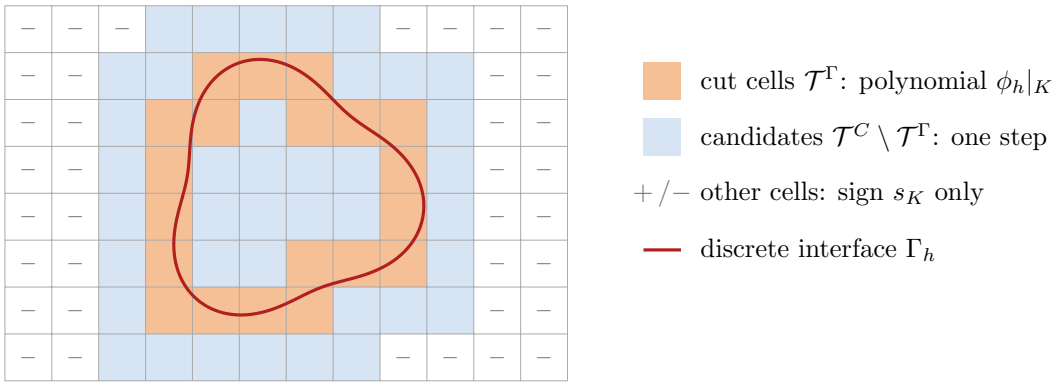


Figure 1: Cell sets of the semi-Lagrangian narrow band. The level set is stored only on the cut cells; the candidate cells (cut cells and their vertex neighbors) receive values only during a time step. All other cells know only on which side of the interface they lie.

2 Discrete level set on the cut cells

Cut cells. A cell K belongs to $\mathcal{T}^\Gamma$ if the zero level set of its polynomial lies inside K. This is decided with the discrete quadrature generator of deal.II (Saye's algorithm for implicitly defined domains [5]): K is cut if the generated surface quadrature for $\{\phi_h|_K = 0\} \cap K$ is not empty.

Degrees of freedom. Two hp-DoFHandlers share the active FE indices. The *cut DoFHandler* holds `FE_DGQ`(p) on $\mathcal{T}^\Gamma$ and `FE_Nothing` elsewhere; it stores the level set between time steps and is used by the reinitialization. The *band DoFHandler* holds `FE_DGQ`(p) on all candidate cells $\mathcal{T}^C$; it receives the semi-Lagrangian values. Both are rebuilt whenever the cut cells change.

Cell signs. A cell that is not cut contains no part of Γ_h, so ϕ has a single sign on it. This sign can only change if the interface passes through the cell, and the interface can only reach a cell through the candidate layer. Therefore one integer per cell suffices:

- initialization: $s_K = \operatorname{sign}\phi_0(\boldsymbol{x}_K)$ at the cell center $\boldsymbol{x}_K$,
- update after each step: for every candidate cell that is not cut at t^{n+1}, the sign of the mean of its N nodal values, $s_K = \operatorname{sign}\big(\frac{1}{N}\sum_i \phi_h^{n+1}(\boldsymbol{x}_i)\big)$,
- the signs of the ghost cells are communicated after each update.

3 Semi-Lagrangian advection

The level set satisfies $\partial_t\phi + \boldsymbol{u}\cdot\nabla\phi = 0$, i.e., it is constant along the characteristics $\dot{\boldsymbol{X}}(t) = \boldsymbol{u}(\boldsymbol{X}(t), t)$. The level set at t^{n+1} is therefore the old level set at the foot points of the characteristics.

Foot points. For every node $\boldsymbol{x}_i$ (support point of `FE_DGQ`) of every candidate cell, the characteristic is integrated backward from t^{n+1} to t^n with the classical fourth-order Runge–Kutta method and the step $-\Delta t$:

$$\begin{aligned} &\boldsymbol{k}_1 = \boldsymbol{u}(\boldsymbol{x}_i, t^{n+1}), && \boldsymbol{k}_2 = \boldsymbol{u}\big(\boldsymbol{x}_i - \tfrac{\Delta t}{2}\boldsymbol{k}_1,\, t^{n+\frac12}\big), \\ &\boldsymbol{k}_3 = \boldsymbol{u}\big(\boldsymbol{x}_i - \tfrac{\Delta t}{2}\boldsymbol{k}_2,\, t^{n+\frac12}\big), && \boldsymbol{k}_4 = \boldsymbol{u}\big(\boldsymbol{x}_i - \Delta t\,\boldsymbol{k}_3,\, t^{n}\big), \end{aligned} \qquad \boldsymbol{x}_i^* = \boldsymbol{x}_i - \frac{\Delta t}{6}\left(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\right). \tag{1}$$

Old level set at the foot point. The new nodal value is the old level set at $\boldsymbol{x}_i^*$ (Figure 2):

$$\phi_h^{n+1}(\boldsymbol{x}_i) = \begin{cases} \phi_h^n|_K(\boldsymbol{x}_i^*), & \boldsymbol{x}_i^* \in K \in \mathcal{T}^{\Gamma,n} \quad \text{(cut cell)}, \\ s_K\,|\nabla\phi_h^n(\boldsymbol{y})|\,d(\boldsymbol{x}_i^*, \boldsymbol{y}), & \boldsymbol{x}_i^* \in K \notin \mathcal{T}^{\Gamma,n} \quad \text{(not cut)}. \end{cases} \tag{2}$$

In the second case, $\boldsymbol{y}$ is the closest point of $\boldsymbol{x}_i^*$ on the old discrete interface Γ_h^n. It is computed per candidate cut cell by Newton's method for the optimality conditions of $\min_{\boldsymbol{y}} \frac12|\boldsymbol{x}_i^* - \boldsymbol{y}|^2$ subject to $\phi_h(\boldsymbol{y}) = 0$ (Saye [4]),

$$\boldsymbol{y} - \boldsymbol{x}_i^* + \lambda\nabla\phi_h(\boldsymbol{y}) = \boldsymbol{0}, \qquad \phi_h(\boldsymbol{y}) = 0, \tag{3}$$

starting from the nearest quadrature point of Γ_h^n. The candidate cut cells are those of the nearest quadrature points of the interface (8 in 2D); the one with the smallest distance estimate

$$d(\boldsymbol{x}_i^*, \boldsymbol{y}) = |\boldsymbol{x}_i^* - \boldsymbol{y}| + \frac{|\phi_h(\boldsymbol{y})|}{|\nabla\phi_h(\boldsymbol{y})|} \tag{4}$$

is taken. For a converged closest point, $s_K|\nabla\phi_h(\boldsymbol{y})|\,d$ equals the first-order Taylor value $\phi_h(\boldsymbol{y}) + \nabla\phi_h(\boldsymbol{y})\cdot(\boldsymbol{x}_i^* - \boldsymbol{y})$. At an open end of Γ_h^n, the Taylor value would vanish on the continuation of the tangent, while the distance to the end point does not.

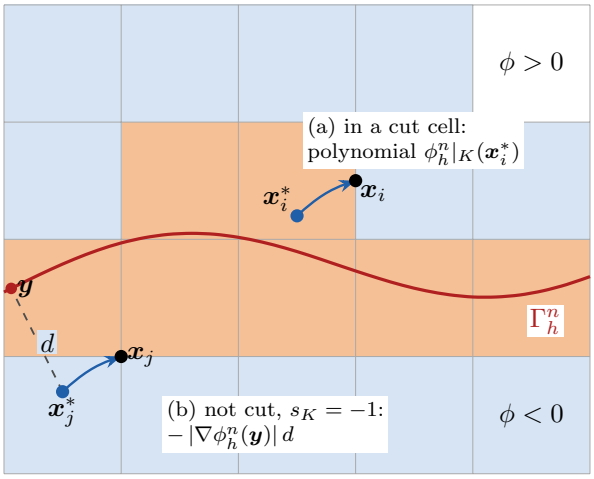


Figure 2: Back-tracing of two nodes of candidate cells (orange: cut cells, blue: candidate cells that are not cut). The arrows are the characteristics from the foot point to the node over one time step. (a) The foot point lies in a cut cell: its polynomial is evaluated. (b) The foot point lies in a cell that is not cut: the scaled distance to the old interface with the stored sign of that cell.

Nodal interpolation and new cut cells. The values (2) are the nodal values of ϕ_h^{n+1} on the band DoFHandler. The new cut cells $\mathcal{T}^{\Gamma,n+1} \subseteq \mathcal{T}^C$ are those candidate cells in which ϕ_h^{n+1} has a zero (Section 2). Their polynomials are copied to the new cut DoFHandler; the values of the other candidate cells are only used to update the signs and are then discarded.

Properties.

- *No extrapolation.* The interface at t^{n+1} can only come from material that was in a cut cell at t^n. A node whose foot point lies in a cell that is not cut always gets the sign of that cell, so no new interface appears there, and an open end of Γ_h (an under-resolved tail) moves with the flow instead of being prolonged.
- *Time step.* There is no stability limit of an explicit DG scheme. The foot points must lie in the candidate cell or its vertex neighbors, i.e., the displacement $|\boldsymbol{x}_i - \boldsymbol{x}_i^*|$ must stay below one cell. The maximum displacement relative to the cell size is printed in each step.
- *Accuracy.* Every step adds an interpolation error of order h^{p+1}, so the error after $T/\Delta t$ steps behaves like $h^{p+1}/\Delta t$, i.e., like h^p for a fixed ratio $\Delta t/h$. Large time steps (displacement of about $0.3h$ to $0.7h$) are more accurate than small ones.
- *Conservation.* The method is not conservative.

4 Elliptic reinitialization on the cut cells

The advected level set loses the signed-distance property. It is restored on the cut cells only, while the position of the interface is kept. Following Basting and Kuzmin [2] and Adams et al. [1], the reinitialized level set minimizes

$$\int_{\Omega_\Gamma} \left(|\nabla\phi| - 1\right)^2 \mathrm{d}x + \frac{\gamma}{2} \int_\Gamma \phi^2 \,\mathrm{d}s, \tag{5}$$

where $\Omega_\Gamma = \bigcup_{K\in\mathcal{T}^\Gamma} K$ is the union of the cut cells and Γ is the zero level set before the reinitialization; the second term enforces the interface position weakly. The nonlinear problem is solved by the fixed-point iteration: find ϕ^{k+1} with

$$a(\phi^{k+1}, v) = l(\phi^k; v) \quad \text{for all } v \in Q_p(\mathcal{T}^\Gamma), \tag{6}$$

Algorithm 1 Semi-Lagrangian evaluation of ϕ_h^{n+1} on the candidate cells

Input: ϕ_h^n on $\mathcal{T}^{\Gamma,n}$, signs s_K, velocity $\boldsymbol{u}$, t^n, Δt
Output: nodal values of ϕ_h^{n+1} on $\mathcal{T}^C = \mathcal{T}^{\Gamma,n} \cup \{\text{vertex neighbors}\}$

gather the cut cells of all processes: affine map, DoF values, quadrature points of Γ_h^n
build search trees of the cut-cell bounding boxes and of the interface quadrature points
for each locally owned cell $C \in \mathcal{T}^C$ **do**
  for each node $\boldsymbol{x}_i$ of C **do**
    $\boldsymbol{x}_i^* \leftarrow$ RK4 backward along $\boldsymbol{u}$ from t^{n+1} to t^n
    if $\boldsymbol{x}_i^* \in K$ for a cut cell K (box tree, then reference coordinates) **then**
      $\phi_h^{n+1}(\boldsymbol{x}_i) \leftarrow \phi_h^n|_K(\boldsymbol{x}_i^*)$
    else
      $\boldsymbol{y} \leftarrow$ closest point on Γ_h^n (Newton in the cut cells of the nearest interface points)
      $K \leftarrow$ cell containing $\boldsymbol{x}_i^*$ among C and its vertex neighbors
      if K found **then**
        $\phi_h^{n+1}(\boldsymbol{x}_i) \leftarrow s_K \, |\nabla \phi_h^n(\boldsymbol{y})| \, d(\boldsymbol{x}_i^*, \boldsymbol{y})$
      else
        $\phi_h^{n+1}(\boldsymbol{x}_i) \leftarrow \operatorname{sign}\big(\phi_h^n(\boldsymbol{y}) + \nabla \phi_h^n(\boldsymbol{y}) \cdot (\boldsymbol{x}_i^* - \boldsymbol{y})\big) \, |\nabla \phi_h^n(\boldsymbol{y})| \, d(\boldsymbol{x}_i^*, \boldsymbol{y})$ ▷ reported
      end if
    end if
  end for
end for

with the SIPG bilinear form on the faces $\mathcal{F}_\Gamma$ between two cut cells

$$a(\phi, v) = (\nabla\phi, \nabla v)_{\Omega_\Gamma} - (\{\nabla\phi\} \cdot \boldsymbol{n}, [v])_{\mathcal{F}_\Gamma} - ([\phi], \{\nabla v\} \cdot \boldsymbol{n})_{\mathcal{F}_\Gamma} + \sigma([\phi], [v])_{\mathcal{F}_\Gamma} + \gamma(\phi, v)_\Gamma, \quad (7)$$

the linear form

$$l(\phi^k; v) = \big(g(|\nabla\phi^k|)\nabla\phi^k, \nabla v\big)_{\Omega_\Gamma} - \big(\{g(|\nabla\phi^k|)\nabla\phi^k\} \cdot \boldsymbol{n}, [v]\big)_{\mathcal{F}_\Gamma}, \qquad g(s) = \begin{cases} 1/s, & s > 1, \\ 2 - s, & s \le 1, \end{cases} \quad (8)$$

and the interior penalty $\sigma = c_{\text{ip}} \, p^3 / h_{\min}$ (default $c_{\text{ip}} = 10$). The faces between a cut cell and a cell that is not cut, and the domain boundary, are part of the boundary of Ω_Γ, where the natural boundary condition of the minimization problem holds; no face integrals are evaluated there (Figure 3). Only the cells and faces of Ω_Γ are visited, so the cost scales with the size of the interface. The linear systems are solved matrix-free with CG.

The reinitialization is triggered every f-th step (parameter `reinitialization frequency`) or when the eikonal error on the cut cells, $\big(\int_{\Omega_\Gamma} (|\nabla\phi_h| - 1)^2 \, \mathrm{d}x / |\Omega_\Gamma|\big)^{1/2}$, exceeds a threshold (`reinitialization eikonal threshold`).

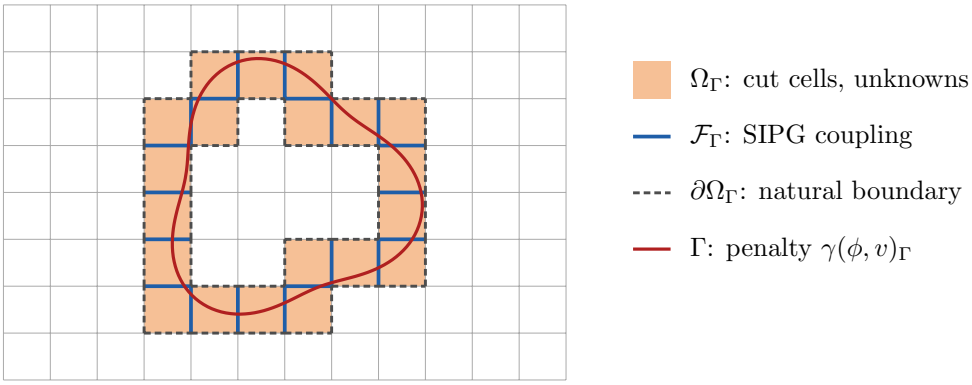


Figure 3: Domain of the elliptic reinitialization: only the cut cells carry unknowns. The cut cells are coupled by SIPG terms across their common faces; at the faces to cells that are not cut, the natural boundary condition applies. The interface position is enforced by a penalty on Γ.

Algorithm 2 Elliptic reinitialization on the cut cells

Input: ϕ_h on $\mathcal{T}^\Gamma$
Output: reinitialized ϕ_h on $\mathcal{T}^\Gamma$, same zero level set (weakly)
1: $\Gamma \leftarrow \{\phi_h = 0\}$; surface quadrature on the cut cells
2: $\phi^0 \leftarrow \phi_h, \quad k \leftarrow 0$
3: **repeat**
4: assemble $l(\phi^k; \cdot)$ on Ω_Γ and $\mathcal{F}_\Gamma$
5: solve $a(\phi^{k+1}, v) = l(\phi^k; v)$ for all v ▷ matrix-free CG
6: $k \leftarrow k + 1$
7: **until** residual $< 10^{-5}$ and $\|\phi^k - \phi^{k-1}\| < 10^{-5}$
8: $\phi_h \leftarrow \phi^k$

5 The complete time step

The reinitialization is the last stage of a time step. It acts on the cut cells found from the advected level set, and the next semi-Lagrangian step traces back into exactly these reinitialized polynomials. The signs need no reinitialization: they only state on which side a cell lies, and the reinitialization keeps the interface in place. Figure 4 shows the flow of one step and Algorithm 3 the complete method.

6 Comparison with the DG band advection

The alternative in `mp-narrow-band` (`narrow band/type = extension`) advects the level set with an upwind DG scheme in non-conservative form on a band of the cut cells and their face neighbors $\mathcal{T}^B$: find $\phi_h \in Q_p(\mathcal{T}^B)$ with

$$\sum_{K \in \mathcal{T}^B} \left[(\partial_t \phi_h + \boldsymbol{u} \cdot \nabla \phi_h, v)_K - \int_{\partial K^-} (\boldsymbol{u} \cdot \boldsymbol{n}_K) \, (\phi_h^{\mathrm{int}} - \phi_h^{\mathrm{ext}}) \, v \, \mathrm{d}s \right] = 0, \tag{9}$$

with the inflow boundary ∂K^- and $\phi_h^{\mathrm{ext}} = \phi_h^{\mathrm{int}}$ at the outer boundary of the band, integrated with a third-order SSP Runge–Kutta method. Before each step, the level set of the neighbor cells is reconstructed by an extension from the cut cells (ghost-penalty or closest-point extension). The reinitialization is the same as in Section 4.

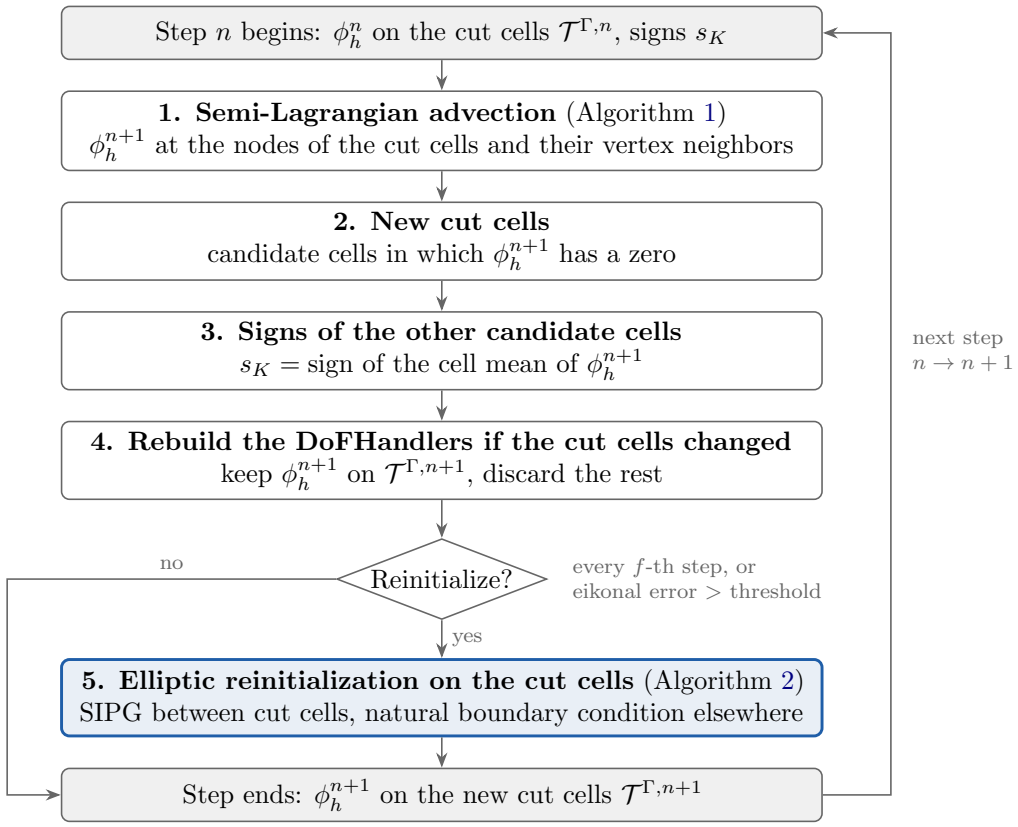


Figure 4: One time step of the semi-Lagrangian narrow band. The reinitialization closes the step and acts on the new cut cells only.

Algorithm 3 Semi-Lagrangian narrow-band level set

Input: initial level set ϕ_0, velocity $\boldsymbol{u}$, Δt, reinitialization frequency f

Initialization

$\mathcal{T}^\Gamma \leftarrow$ cells intersected by ϕ_0 (mesh classifier, then the discrete check of Section 2)

$\phi_h \leftarrow$ interpolation of ϕ_0 on $\mathcal{T}^\Gamma$; $\quad s_K \leftarrow \operatorname{sign}\phi_0(\boldsymbol{x}_K)$ for all cells

optionally $\phi_h \leftarrow$ Reinitialize(ϕ_h, $\mathcal{T}^\Gamma$)

Time loop

for $n = 0, 1, \ldots$ **do**

  $\mathcal{T}^C \leftarrow \mathcal{T}^\Gamma \cup \{K : K$ shares a vertex with a cut cell$\}$

  $\phi_h^{n+1} \leftarrow$ semi-Lagrangian evaluation on $\mathcal{T}^C$ (Algorithm 1)

  $\mathcal{T}^{\Gamma,n+1} \leftarrow \{K \in \mathcal{T}^C : \phi_h^{n+1}|_K$ has a zero in $K\}$

  for $K \in \mathcal{T}^C \setminus \mathcal{T}^{\Gamma,n+1}$ **do**

    $s_K \leftarrow$ sign of the mean of the nodal values of $\phi_h^{n+1}|_K$

  end for

  exchange s_K with the ghost cells

  if $\mathcal{T}^{\Gamma,n+1} \neq \mathcal{T}^\Gamma$ **then**

    rebuild the cut and band DoFHandlers

  end if

  $\phi_h \leftarrow \phi_h^{n+1}|_{\mathcal{T}^{\Gamma,n+1}}$; $\quad \mathcal{T}^\Gamma \leftarrow \mathcal{T}^{\Gamma,n+1}$

  if $(n+1) \bmod f = 0$ **or** eikonal error > threshold **then**

    $\phi_h \leftarrow$ Reinitialize(ϕ_h, $\mathcal{T}^\Gamma$) ▷ Algorithm 2

  end if

end for

Table 1: DG band advection and semi-Lagrangian narrow band.

	DG band (`extension`)	semi-Lagrangian (`semi_lagrangian`)
Stored level set	cut cells and face neighbors	cut cells only (plus one sign per cell)
Data outside the cut cells	polynomial extension in every step	none; distance with the sign of the cell
Time step	explicit DG limit, CFL $\sim 1/(2p+1)$	displacement below one cell
Order (resolved)	$p+1$ (observed 2.2–2.6 for $p=2$)	p (observed 1.9 for $p=2$)
Under-resolved tails	extrapolation prolongs open ends, spurious interfaces	tail breaks up, no spurious interfaces
Extra cost	extension (global linear solve on the band)	closest-point search per node outside the cut cells

7 Results

All results use $p=2$, the elliptic reinitialization on the cut cells with the interface penalty $\gamma = 10^4$, and the interface error $e_\Gamma = \left(\int_{\Gamma_h} d_{\text{ex}}^2 \, \mathrm{d}s / |\Gamma_h| \right)^{1/2}$ with the exact signed distance d_{ex} at the final time.

Rotating circle. A circle of radius 0.25 centered at $(0, 0.5)$ rotates once about the origin in $\Omega = [-1,1]^2$ (angular velocity 2π, $t \in [0,1]$). Table 2 and Figure 5 compare the DG band advection (ghost-penalty extension, reinitialization every 10 steps) with the semi-Lagrangian method at the same and at a four times larger time step.

Table 2: Rotating circle, interface error e_Γ after one revolution (EOC: experimental order of convergence). DG band and SL with Δt: $\Delta t = 0.004, 0.002, 0.001$, reinitialization every 10 steps. SL with $4\Delta t$: reinitialization every 3, 3, 2 steps.

cells	h	DG band, Δt		SL, Δt		SL, $4\Delta t$	
		e_Γ	EOC	e_Γ	EOC	e_Γ	EOC
16^2	0.125	$1.32 \cdot 10^{-3}$	–	$7.88 \cdot 10^{-3}$	–	$3.30 \cdot 10^{-3}$	–
32^2	0.0625	$2.17 \cdot 10^{-4}$	2.61	$2.19 \cdot 10^{-3}$	1.85	$8.90 \cdot 10^{-4}$	1.89
64^2	0.03125	$4.86 \cdot 10^{-5}$	2.16	$5.53 \cdot 10^{-4}$	1.98	$2.38 \cdot 10^{-4}$	1.91

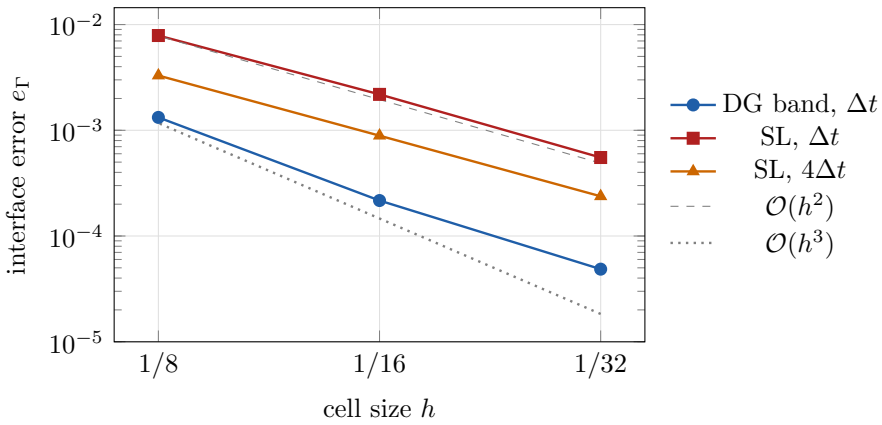


Figure 5: Rotating circle: convergence of the interface error. The semi-Lagrangian method converges with second order; the larger time step reduces the error by a factor of about 2.4.

Single vortex. A circle of radius 0.15 centered at $(0.5, 0.75)$ in $\Omega = [0,1]^2$ is stretched by the time-reversed vortex $\boldsymbol{u} = \cos(\pi t/T)\left(\sin(2\pi y)\sin^2(\pi x), -\sin(2\pi x)\sin^2(\pi y)\right)$ (LeVeque [3]) and

returns to its initial shape at $t = T$. Table 3 lists the results; the area error is the relative change of the enclosed area between $t = 0$ and $t = T$, and the circularity is $2\sqrt{\pi A}/|\Gamma_h|$.

Table 3: Single vortex at $t = T$ (f: reinitialization frequency). DG band: ghost-penalty extension.

method	cells	T	Δt	f	e_Γ	area error	circularity
DG band	32^2	2	0.002	10	fails at $t \approx 1.22$ (spurious interfaces)		
DG band	64^2	2	0.001	10	$1.58 \cdot 10^{-4}$	$-0.06\,\%$	1.000
SL	32^2	2	0.01	2	$1.56 \cdot 10^{-3}$	$+0.10\,\%$	0.998
SL	32^2	2	0.002	10	degrades after $t \approx 1.6$		
SL	64^2	2	0.005	2	$2.25 \cdot 10^{-4}$	$-0.02\,\%$	1.000
SL	32^2	4	0.01	2	$1.31 \cdot 10^{-2}$	$-1.1\,\%$	0.91
SL	64^2	4	0.005	2	$5.59 \cdot 10^{-2}$	$+0.75\,\%$	0.69

For $T = 4$, the tail of the spiral becomes thinner than a cell at both resolutions (Figure 6). With the semi-Lagrangian method, the under-resolved tail breaks into pieces (32^2) or pinches off as a droplet (64^2), which does not merge back; this droplet dominates the interface error and the circularity at 64^2. No spurious interfaces appear, and the number of cut cells stays bounded. On the coarse mesh the result is sensitive to small perturbations: with 4 instead of 2 MPI processes (different ILU preconditioner of the reinitialization), the area error at $T = 4$ is $-6\,\%$ instead of $-1.1\,\%$.

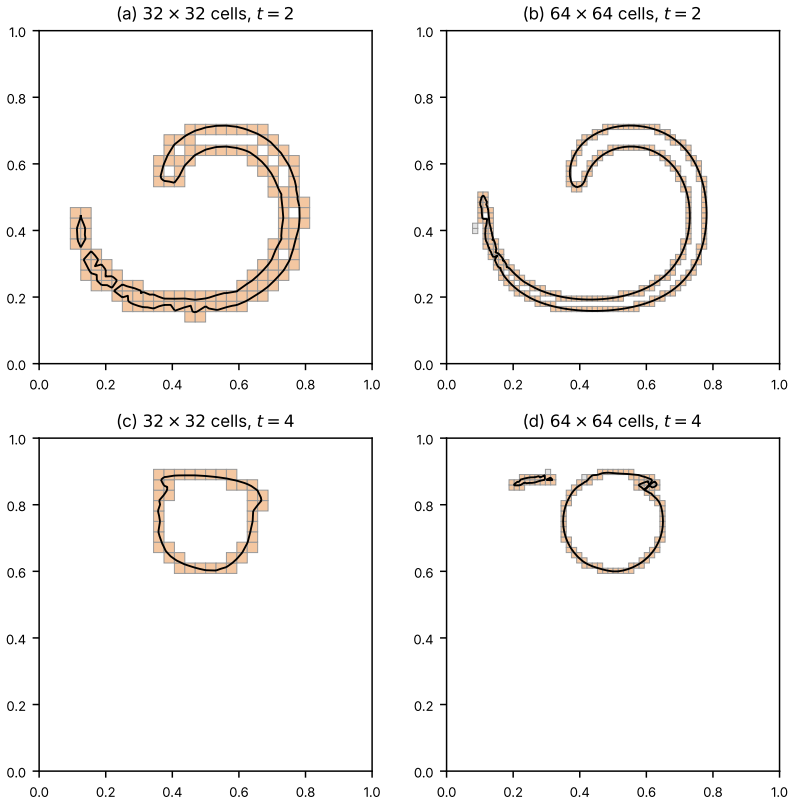


Figure 6: Single vortex with $T = 4$, semi-Lagrangian narrow band: cut cells (orange) and zero level set at maximum stretching ($t = 2$) and at the end ($t = 4$).

8 Parameters and limitations

Table 4: Recommended parameters of the semi-Lagrangian narrow band.

parameter	value
`narrow band/type`	`semi_lagrangian`
`time stepping/time step size`	max. displacement$/h \approx 0.3$–0.7
`narrow band/reinitialization frequency`	2
`reinitialization/type`	`elliptic`
`reinitialization/elliptic/intersected cells only`	`true`
`reinitialization/elliptic/penalty parameter`	10^4

- The displacement per step must stay below one cell; foot points outside the vertex neighborhood are reported as *not located* and get the sign of the Taylor value.
- Small time steps accumulate interpolation errors. When coupled to a flow solver with small time steps, the level set should be advected only every k-th flow step.
- The velocity is evaluated at arbitrary points and intermediate times. With a discrete velocity field, this requires point evaluation across processes and interpolation in time.
- All cut cells are gathered on every process for the closest-point search. This is acceptable for 2D and moderate 3D interfaces; large 3D runs need a local search among neighboring processes.
- The implementation requires parallelepiped cells, tensor-product nodal elements (`FE_DGQ`) and meshes without hanging nodes. The method is not conservative.

References

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